

Revisiting Coronary Artery Anomalies

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See the slide presentation [here](#).

Coronary artery anomalies (CAAs) are uncommon anatomic variations that have the potential to be life-threatening. They occur in about 1% of the general population but are often not diagnosed or even suspected during a person's lifetime due to the lack or nonspecificity of symptoms and limitations of routine diagnostic tests.

The clinical impact of CAAs, a heterogeneous group of congenital disorders related to varying pathophysiologic mechanisms, depends on the relationship between the coronary artery's function and its ability to provide adequate blood flow to the myocardium. While some forms of CAAs are benign and mostly diagnosed incidentally, others can cause severe symptoms and even sudden cardiac death. Therefore, the assessment of CAAs and prediction of their varied natural prognoses can be challenging when solely relying on clinical examination.

The increasing adoption of coronary CT angiography (CTA) as a valuable diagnostic modality for patients with chest pain has led to a higher incidence of detecting CAAs in routine clinical practice. In this context, coronary CTA has emerged as a pivotal imaging tool for detecting and characterizing CAAs by providing optimal in vivo anatomic details on cross-sectional and multiplanar reconstruction images, with some advantages over invasive techniques. Conventionally, catheter angiography for evaluation of anomalous vessels may not have been able to adequately define the pre-

cise course of the vessel due to the complex three-dimensional geometry being visualized in two dimensions fluoroscopically. Further, coronary CTA can delineate different anatomic high-risk features that may lead to physiologic high-risk consequences, such as interarterial course, slitlike ostium, acute take-off angle, intramural course, elliptic vessel shape, and proximal vessel narrowing.

More recently, the role of coronary CTA in the assessment of CAAs has been further cemented by the latest technological improvements in CT scanners. These advancements include dual-source technology, faster gantry rotation times, new cardiac imaging modalities, and the contemporary introduction of machine learning–based fractional flow reserve derived from CT.

In this slide presentation, a practical radiologic approach is proposed to facilitate understanding of CAAs and their common and uncommon imaging features through the presentation of sample cases and an assortment of three-dimensional reconstructions. Of note, a basic principle of coronary classification should be that the nature and name of a specific coronary artery are assigned, not according to the site of origin or proximal course, but according to the dependent territory. The authors provide a didactic approach to a proposed comprehensive classification scheme for coronary artery anomalies, based on four categories: (a) origination and course, (b) intrinsic anatomy, (c) termination, and (d) anastomotic vessels (Figure).



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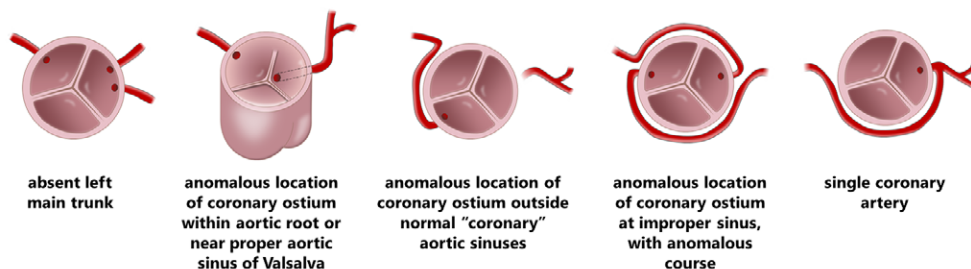
Abbreviations: CAA = coronary artery anomaly, CTA = CT angiography

Content Codes: CA, CT

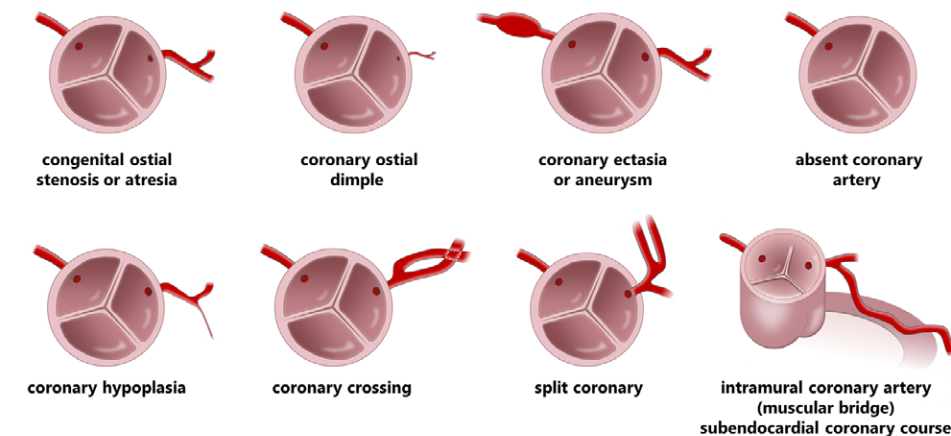
TEACHING POINTS

- CAAs are rare and potentially life-threatening variants, rarely diagnosed or even suspected during life, with many different entities and natural prognoses.
- Coronary CTA has emerged as a cornerstone for noninvasive identification and characterization of CAAs and is useful for catheter-based interventional therapy or surgical planning.
- Coronary CTA can delineate different anatomic high-risk features that may lead to physiologic high-risk consequences, such as interarterial course, slitlike ostium, acute take-off angle, intramural course, elliptic vessel shape, and proximal vessel narrowing.

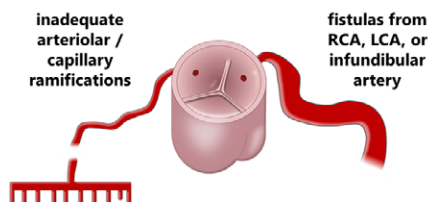
1 | Anomalies of origination and course



2 | Anomalies of intrinsic coronary arterial anatomy



3 | Anomalies of coronary termination



4 | Anomalous anastomotic vessels

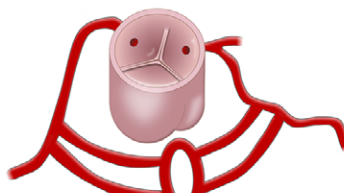


Figure. Classification of CAAs based on four categories: (1) origination and course, (2) intrinsic anatomy, (3) termination, and (4) anastomotic vessels.

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Suggested Readings

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